

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pev}(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

Near-field limit PDE

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K^* \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

Notation:

$$g_\tau(x, \theta) = g(x) + \tau v(\theta)$$

$$\circ \quad f * g(x) = \int_{\mathbb{T}^2} f(x-y)g(y)dy$$

$$\circ \rho(x) = \int_0^{2\pi} f(x, \theta) d\theta$$

Antanalysis

Ant analysis

$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$

- Mean-field limit PDE

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

- Notation:

- $g_\tau(x, \theta) = g(x + \tau v(\theta))$
- $f * g(x) = \int_{\mathbb{T}^2} f(x - y) g(y) dy$
- $\rho(x) = \int_0^{2\pi} f(x, \theta) d\theta$

Ant analysis

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

(dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):

- PDE always globally-in-time well-posed for singular Newtonian ∇K and $\sigma_x > 0$
- No blow-up in infinite time (Alikakos, CPDE, 1979)